WATCH & WONDER

Paper bridges: the power of a fold

GUIDED NOTES

Name: _____ Date: _____

Watch how a simple sheet of paper becomes a bridge. Listen for the changes that make a fair test. Pause when you need time to write.

Fill in the blanks

Word bank: gap / shape

1. A bridge spans a ___ and supports a load.

2. Change only the ___ of the paper to make the comparison fair.

While you watch

1. What are the two bridge designs being compared?

2. Name two things that should stay the same during the comparison.

3. Why do the builders repeat the test three times?

After you watch

1. What would you change in a second bridge design? What would you keep the same to compare it fairly?

2. What surprised you about changing the shape of the paper? Use one observation to explain your thinking.

Try it yourself

Suggested extension - apply an idea from the video. Read the whole activity with an adult before starting.

1. Suggested extension: design your own paper bridge.

Goal: Find a paper shape that spans a gap and supports a small load.
Materials: two books, two equal sheets of paper, and several identical coins. Ask an adult before testing, and keep coins away from younger children.

1. Set the books a short distance apart.
2. Test a flat paper bridge. Add one coin at a time.
3. Fold a second sheet into a shape of your choice. Keep the gap the same.
4. Repeat each test three times and record the coin counts.

My design held _____, _____, and _____ coins.
What feature would I improve next, and why?

Words to know

1. In your own words, what does load mean in this experiment?

Sketch & explain

1. Draw your bridge from the side. Label the gap, the load, and the folded paper. Explain how your sketch connects to the test.